

Addition of açai (Euterpe oleracea) to cigarettes has a protective effect against emphysema in mice.

Chronic inhalation of cigarette smoke (CS) induces emphysema by the damage contributed by oxidative stress during inhalation of CS. Ingestion of açai fruits (*Euterpe oleracea*) in animals has both antioxidant and anti-inflammatory effects. This study compared lung damage in mice induced by chronic (60-day) inhalation of regular CS and smoke from cigarettes containing 100mg of hydroalcoholic extract of açai berry stone (CS + A). Sham smoke-exposed mice served as the control group. Mice were sacrificed on day 60, bronchoalveolar lavage was performed, and the lungs were removed for histological and biochemical analyses. Histopathological investigation showed enlargement of alveolar space in CS mice compared to CS + A and control mice. The increase in leukocytes in the CS group was higher than the increase observed in the CS + A group. Oxidative stress, as evaluated by antioxidant enzyme activities, myeloperoxidase, glutathione, and 4-hydroxynonenal, was reduced in mice exposed to CS+A versus CS. Macrophage and neutrophil elastase levels were reduced in mice exposed to CS + A versus CS. Thus, the presence of açai extract in cigarettes had a protective effect against emphysema in mice, probably by reducing oxidative and inflammatory reactions. These results raise the possibility that addition of açai extract to normal cigarettes could reduce their harmful effects. Copyright © 2010 Elsevier Ltd. All rights reserved.